

RAG Pipeline Architecture Guide

Chapter 1: Retrieval Fundamentals

Retrieval-Augmented Generation (RAG) connects large language models to external knowledge bases. The core idea is simple: retrieve relevant information from a knowledge base and use it to generate a response.

The embedding model converts text into dense vectors. BAAI/bge-small-en-v1.5 produces 384-dimensional vectors and has a maximum context length of 1024 tokens.

BM25 is a probabilistic ranking function from classical information retrieval. It scores documents by term frequency and inverse document frequency.

Chapter 2: Hybrid Search

Hybrid search combines vector similarity with BM25 keyword matching. The merge uses Reciprocal Rank Fusion (RRF), a formula published by Herbolav et al. (2016).

The formula is: $RRF(d) = \sum (1 / (k + rank_i(d)))$ for each ranked list i .

Chapter 3: Re-Ranking

Cross-encoder models process query and document together, producing more accurate relevance scores than bi-encoders. The trade-off is higher computational cost.

Chapter 4: Evaluation

Recall@K measures what fraction of relevant documents appear in the top K results. Mean Reciprocal Rank (MRR) measures the reciprocal of the rank of the first relevant document.

Chapter 5: Production Considerations

ChromaDB stores vectors using HNSW indexing with cosine distance. It persists to disk and supports metadata filtering with operators like `$gt`, `$lt`, `$in`, and `$ne`.

The default rate limit for the Anthropic API depends on your plan tier. Claude Sonnet 4.6 has a context window of 200,000 tokens. Cost per token is \$3.00 for input and \$1.50 for output.

For batch processing, the BM25 index must be reconstructed from the full corpus at initialization time. The `rank-bm25` library does not support incremental updates.

Chapter 6: Safety and Grounding

The system prompt must instruct the model to use **ONLY** the provided context. Without explicit grounding constraints, the model supplements answers with information from its training data.